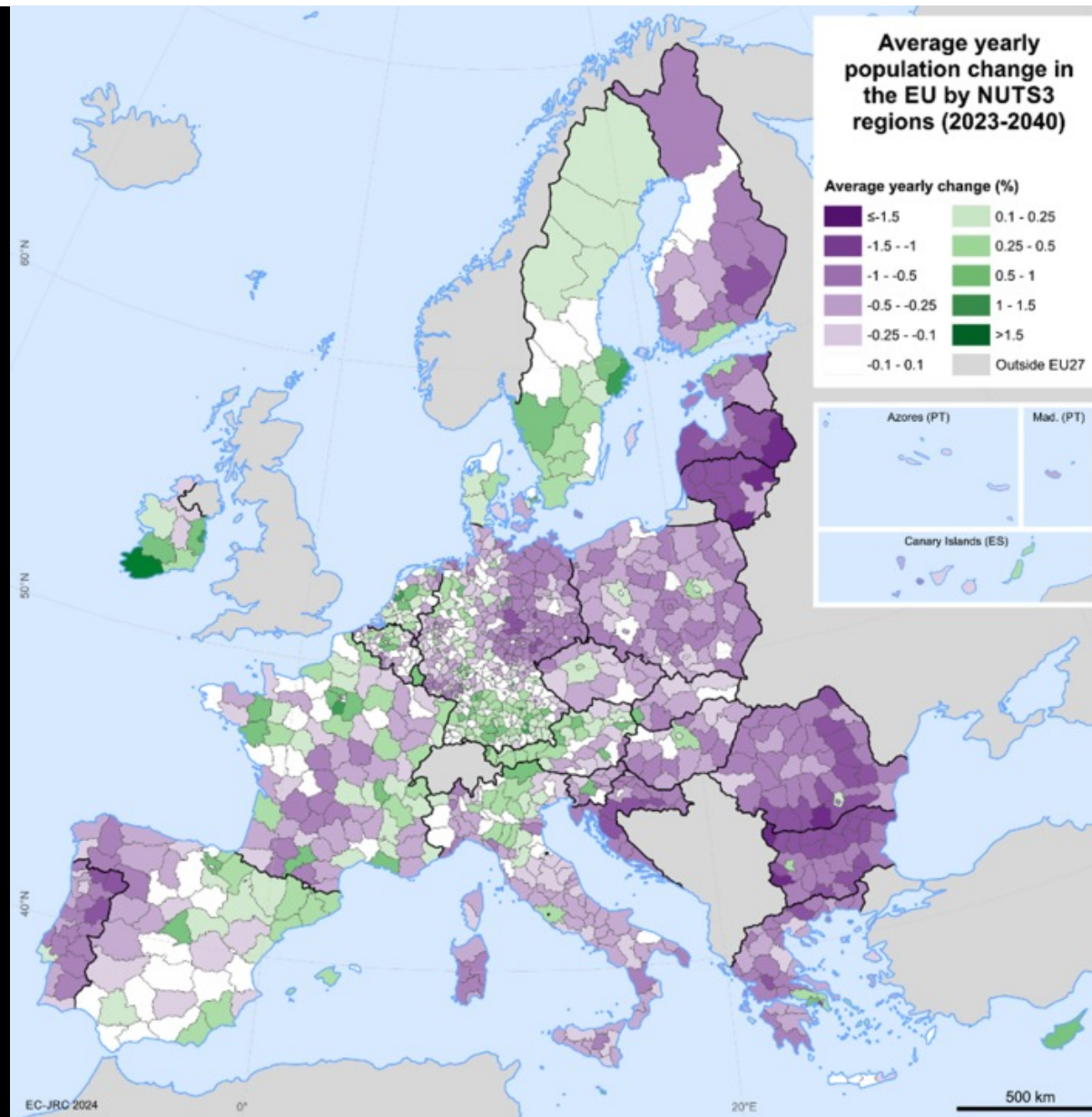


Geo-Processing

GEOG 1026

Department of Geography
National Taiwan University

Curtale et al. (2025)

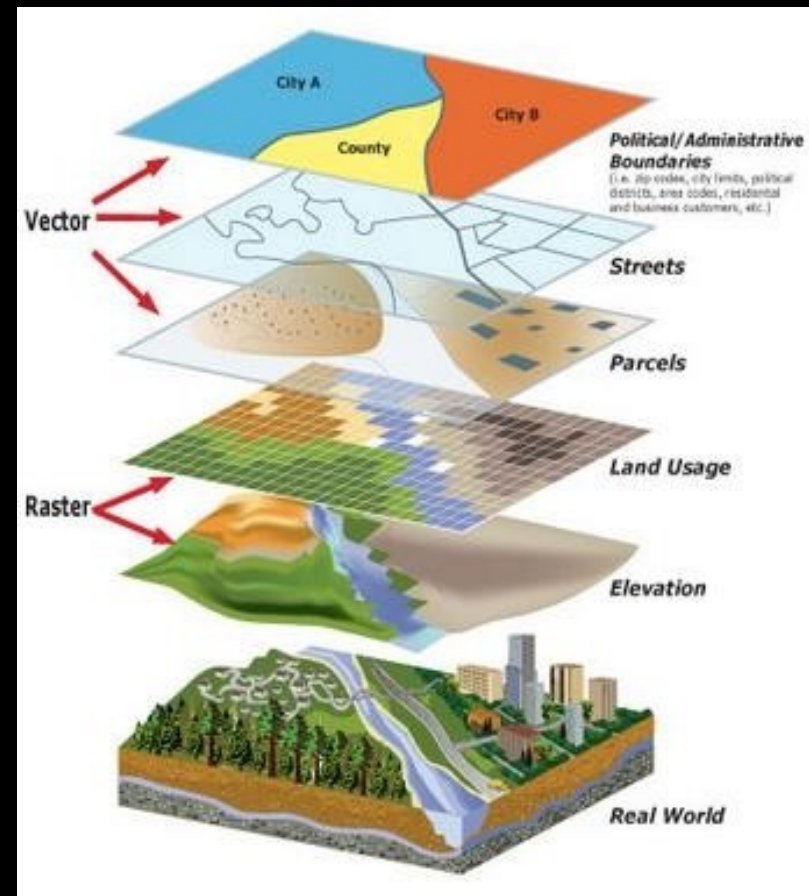


期末報告提案互評

- 將影片共享連結設定為公開
- 至線上測驗的「期末報告提案(影片與計畫書)同儕互評」區進行
- 評分時請略過自己的組別，需要對其他所有組別進行評分
- 若有任何組員未進行評分，則自己的組別無法進入期末口頭報告
- 參考影片與計畫書，考量「創新性」與「題目構想完整性」（各1-5分）兩個面向中同學期待看到期末成果的程度，給予總分（2-10分）的數字來評分
- 同儕互評截止時間為5/15(五)下午2點

Fundamental Geo-Processing and GIS Analysis

- Spatial and attribute query
- Data transformation
- Data management
- Spatial extraction
- Spatial overlay
- Proximity analysis
- Spatial statistics analysis
- Data classification
- Spatial interpolation
- Terrain analysis
- Network analysis



Fundamental Geo-Processing and GIS Analysis

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Fundamental Geo-Processing and GIS Analysis

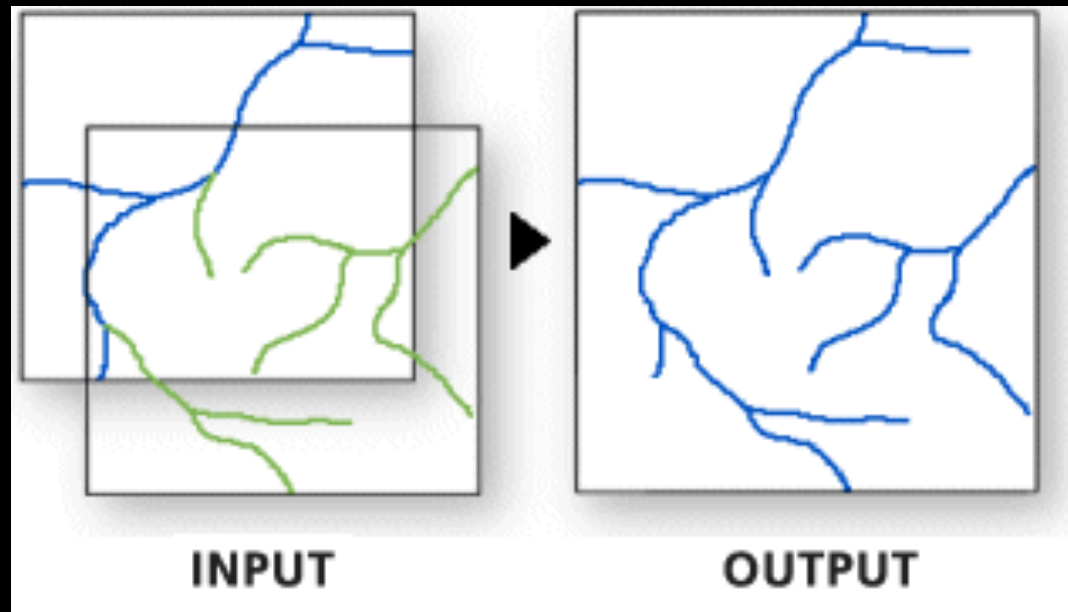
- Spatial and attribute query
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Data Management

- To develop, manage, and maintain feature classes, datasets, layers, and raster data structure.
- Simply manipulating data and do not alter any relationship.
- Common operations:
 - Copy, delete, rename, sort, transfer files
 - Joint/relate
 - Append
 - Merge
 - Dissolve

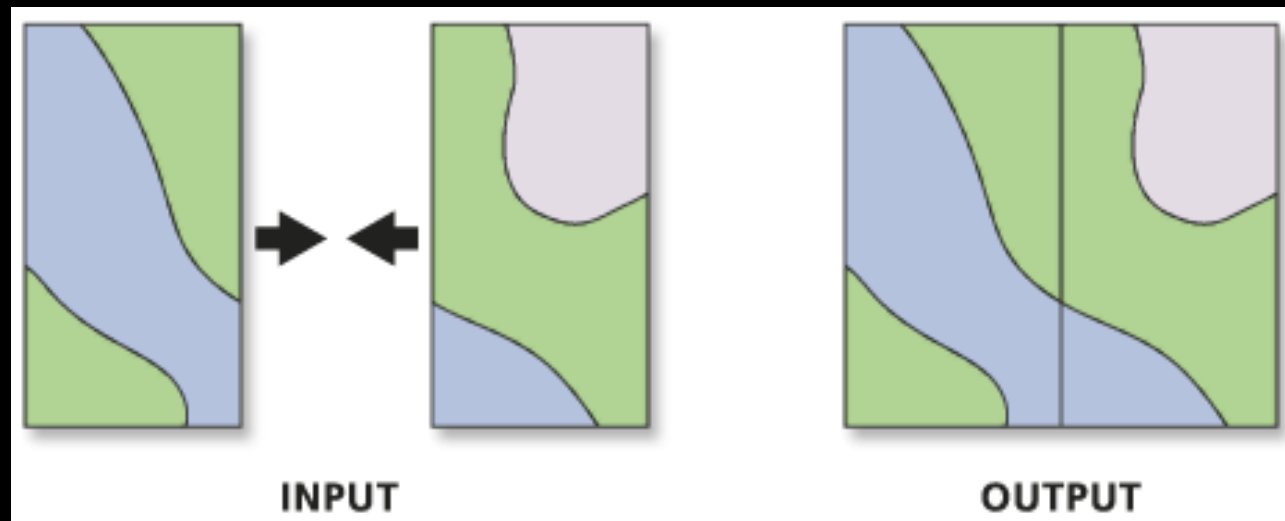
Append (追加)

- Appends to an existing target dataset with multiple input datasets.
- Input datasets can be feature classes, tables, shapefiles, rasters, or annotation or dimension feature classes.



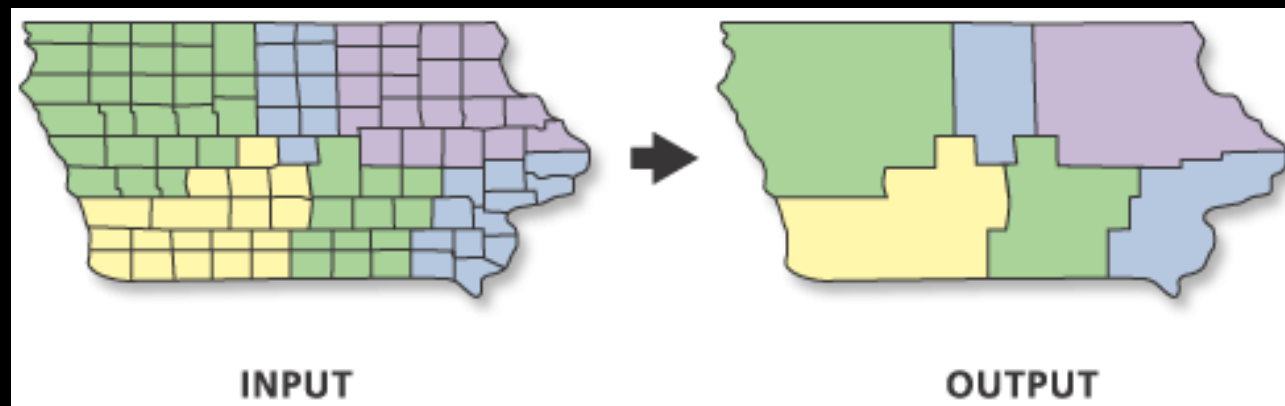
Merge (合并)

- Combine multiple input datasets into a single, new output datasets.
- The tool can combine point, line, or polygon feature classes or tables.



Dissolve (融合)

- Creates a new coverage by merging adjacent polygons, lines, or regions that have the same value for a specified item.
- Aggregates features based on specified attributes.



Fundamental Geo-Processing and GIS Analysis

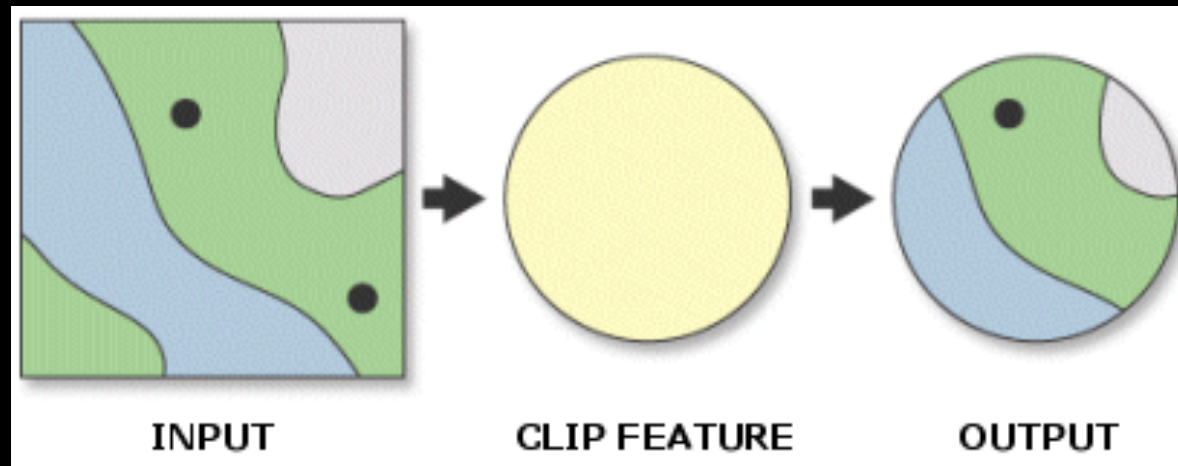
- Spatial and attribute query
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- **Spatial extraction**
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Spatial Extraction (空間擷取)

- Select features and attributes in a feature class or table based on a query or spatial and attribute extraction.
- The output features and attributes are stored in a feature class or table.
- Common operations:
 - Clip
 - Split

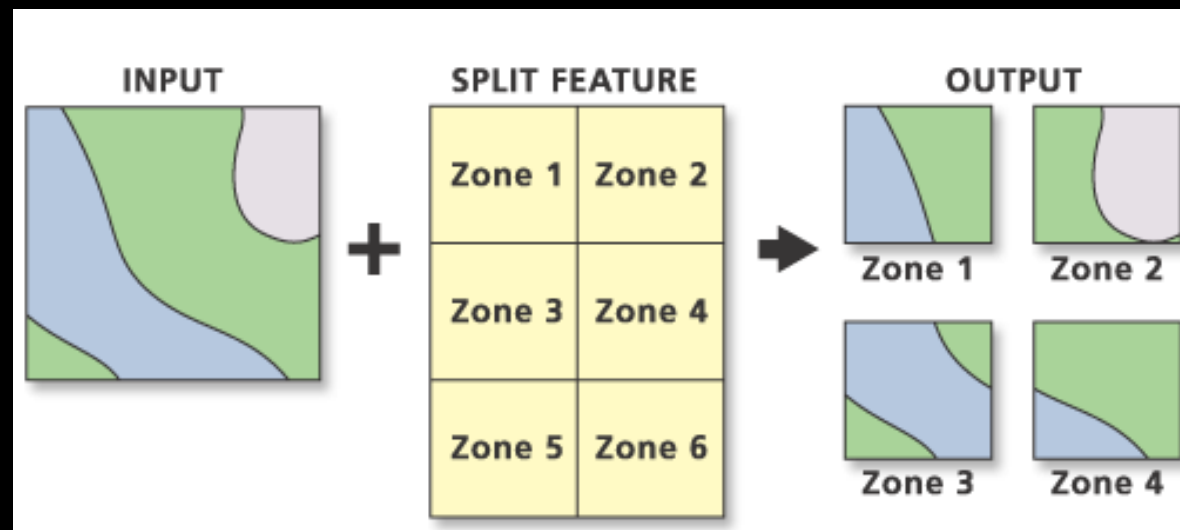
Clip (裁切)

- Extract features or parts of features from a feature class.
- Uses a polygon boundary to cut features and their attribute from a feature class.



Split (分割)

- Splits an input with overlaying features to create a subset of output feature classes.



Fundamental Geo-Processing and GIS Analysis

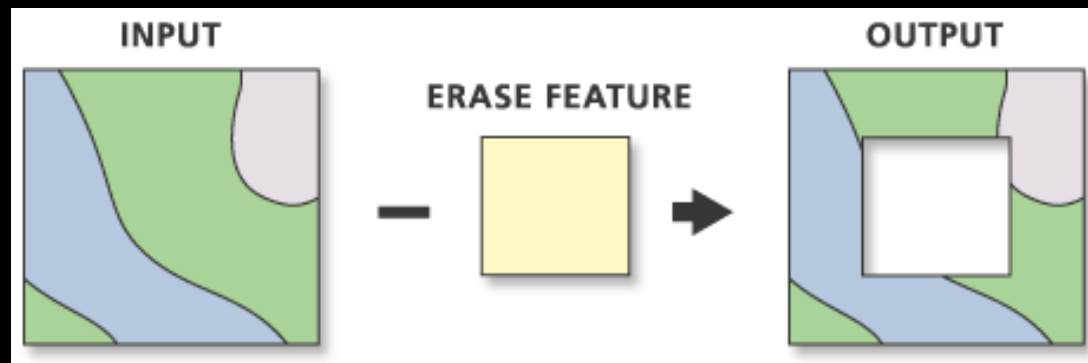
- Spatial and attribute query
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Spatial Overlay (疊圖)

- Spatial overlay functions can combine, erase, modify, or update spatial features, resulting in a new feature class. New information will be created when overlaying sets of features.
- All of overlay operations involve joining two sets of features into a single set of features to **identify spatial relationships** between the input features.
- Common operations:
 - Erase
 - Intersect
 - Union
 - Identity
 - Symmetrical difference
 - Update

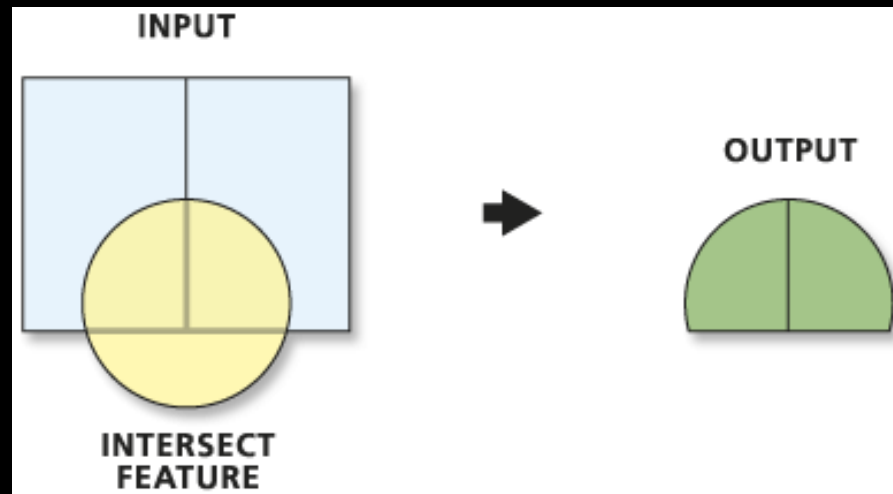
Erase (差集 / 擦除)

- Create a feature class from those features or portions of features outside the erase feature class.
- Only those portions of the Input Features falling outside the boundaries of Erase Features are copied to the Output Feature Class.



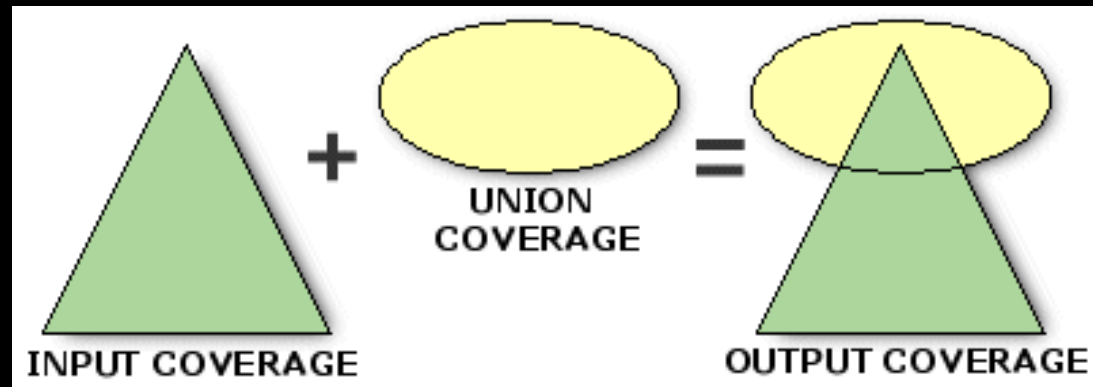
Intersect (交集)

- Creates a new feature from the area where features overlap.
- Features or portions of features which overlap in all layers will be written to the Output Feature Class.



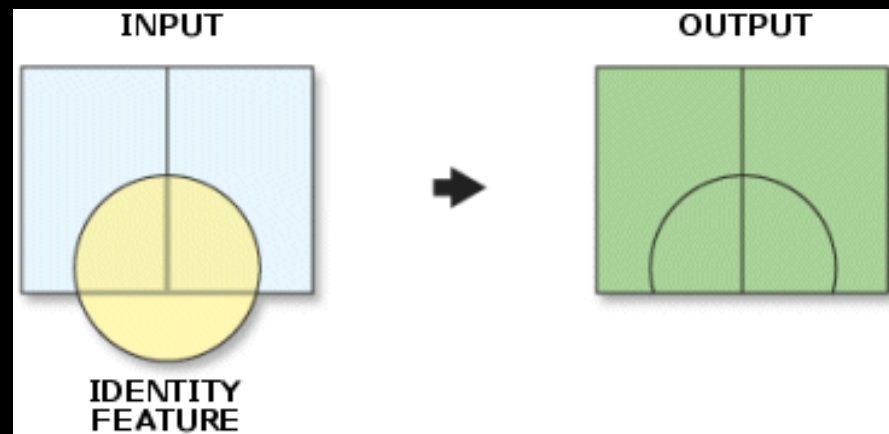
Union (聯集)

- Builds a new feature class by combining the features and attributes of each feature class.
- All features will be written to the Output Feature Class with the attributes from the Input Features.



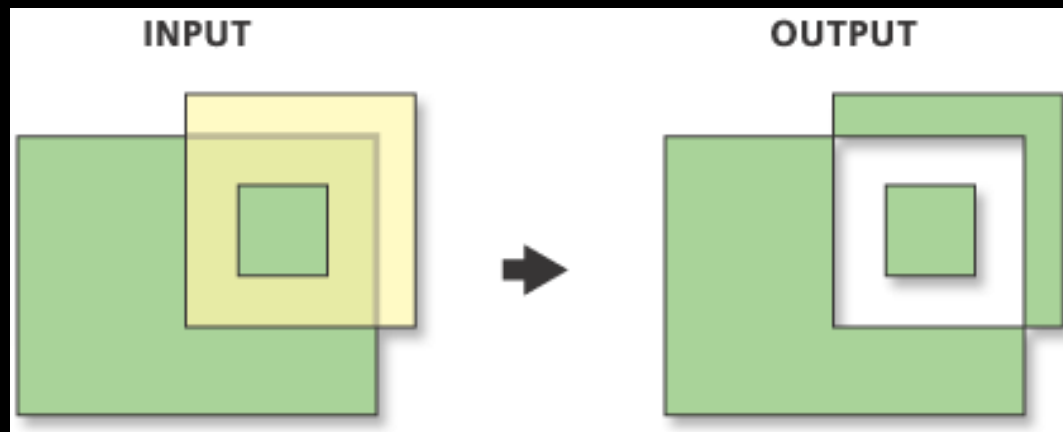
Identity (識別)

- Computes a geometric intersection of the input features and identity features.
- The input features or portions thereof that overlap identify features will get the attributes of those identify features.



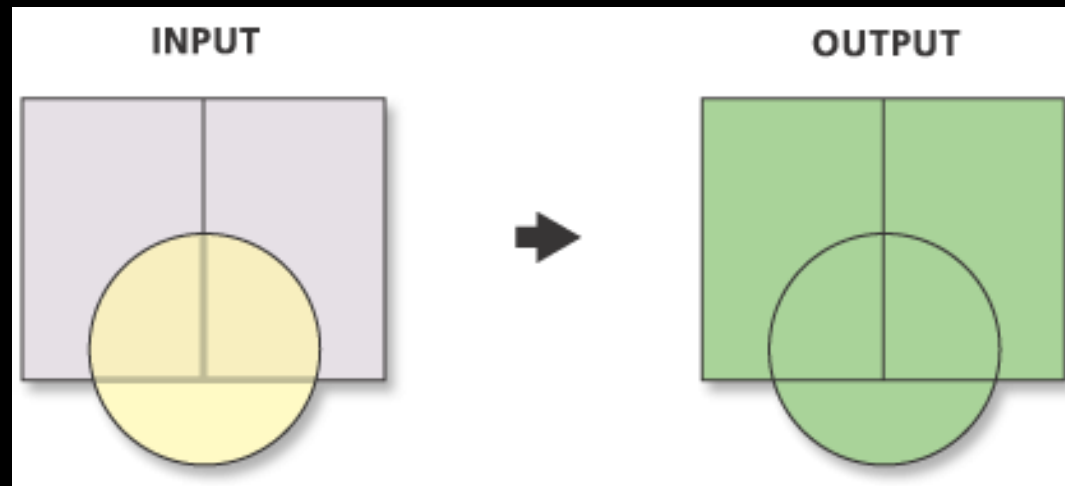
Symmetrical Difference (對稱差)

- Computes a geometric intersection of the input and update features, returning the input features and update features that do not overlap.
- Features or portions of features in the input and update features that do not overlap will be written to the output feature class.

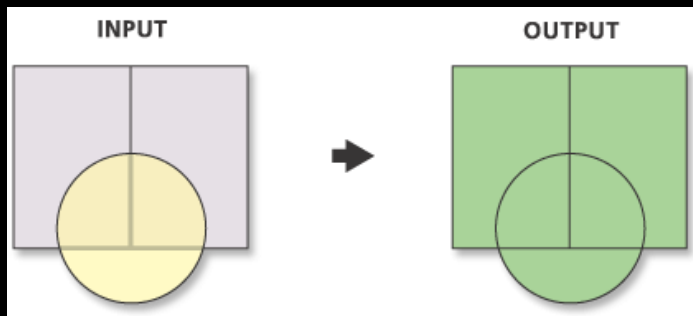
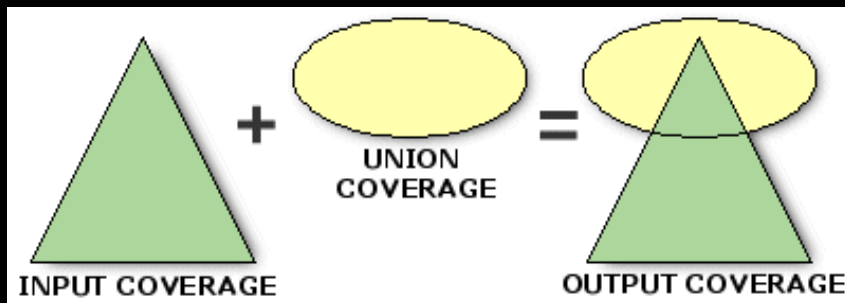
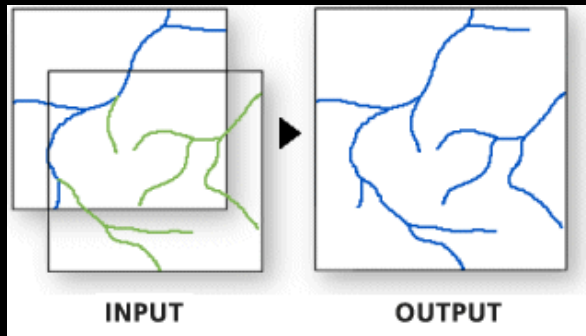


Update (更新)

- Computes a geometric union of the input features.
- All features and their attributes will be written to the output feature class.



Difference between Append, Union, and Update



- Append: only adding data to the original layer
- Union: dissecting the two features to three and building a new feature class to include multiple features.
- Update: replace the data in the overlap area

Fundamental Geo-Processing and GIS Analysis

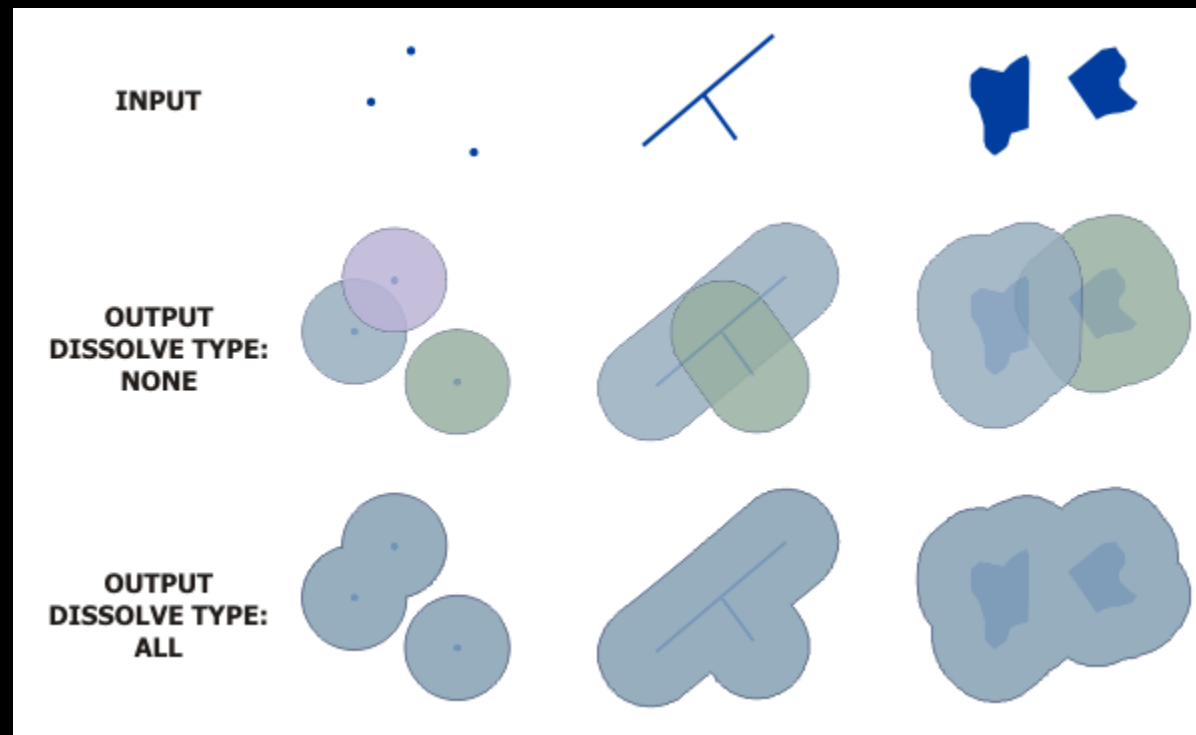
- Spatial and attribute query
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Proximity Analysis (鄰近分析)

- It determined the proximity of features within one or more feature classes or between two feature classes.
- It can identify features that are closet to one another or caluculate the distances between or around them.
- Common operations:
 - Buffer
 - Near

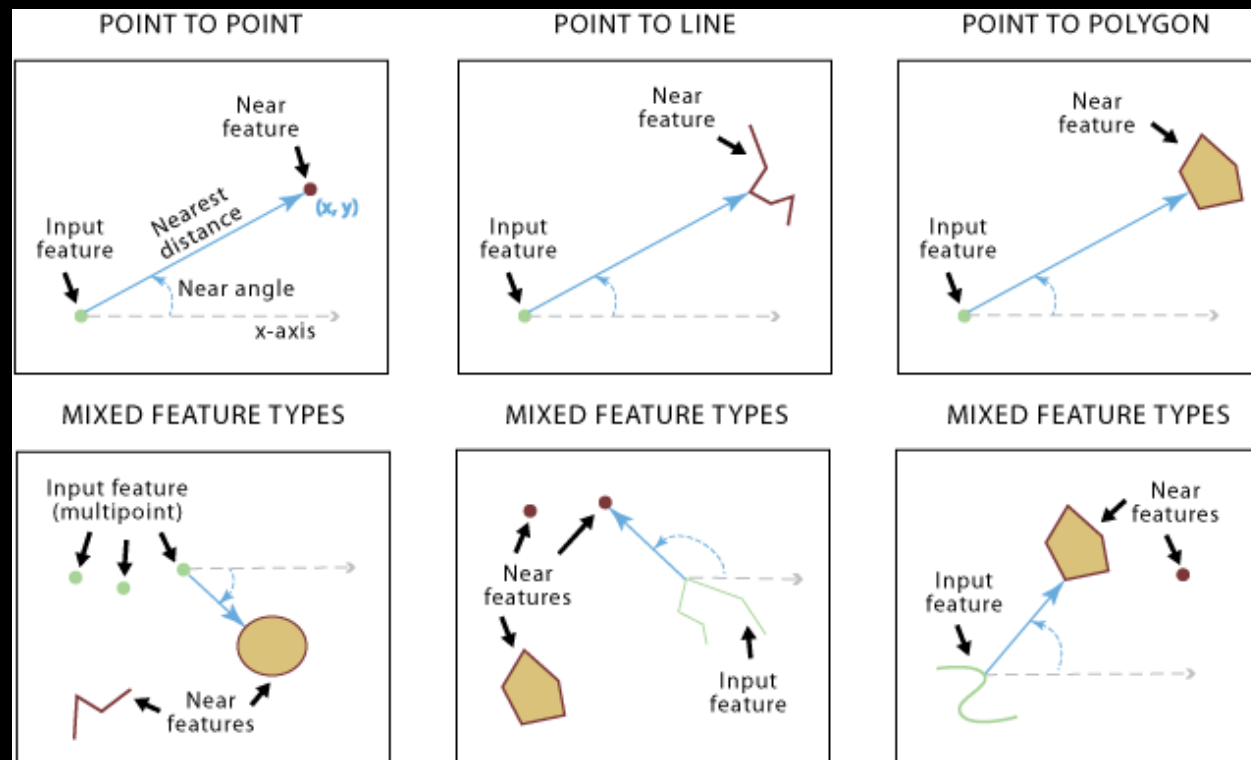
Buffer (環域)

- It creates buffer polygons around input features to a specified distance.



Near (最近距離)

- It calculates distance and additional proximity information between the input features and the closet feature in another layer or feature class.



Fundamental Geo-Processing and GIS Analysis

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Spatial Statistics Analysis (空間統計分析)

- Statistical tools for analyzing spatial distributions, patterns, processes, and relationships.
- Spatial statistics are unique in that they were developed specifically for use with geographic data.
- Common operation:
 - Modeling spatial relationship
 - Spatial autocorrelation
 - Cluster analysis
 - Hotspot analysis

Spatial Autocorrelation (Global Moran's I)

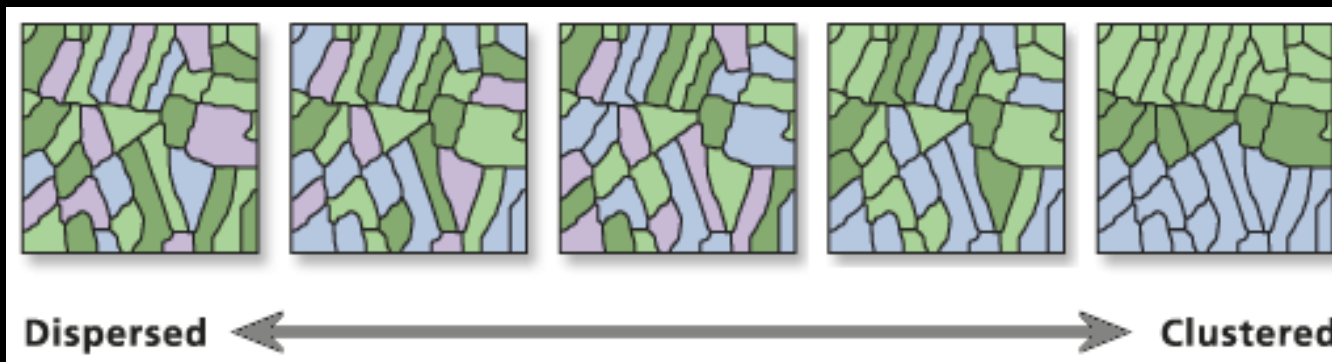
- Measure spatial autocorrelation based on feature locations and attribute values using the Global Moran's I statistic.

To examine if the neighbor cells have
the same tendency of deviation

$$I = \frac{N}{S_0} \frac{\sum_{i=1}^N \sum_{j=1}^N w_{ij} \overbrace{(x_i - \bar{x})}^{\text{Deviation at } X_i} \overbrace{(x_j - \bar{x})}^{\text{Deviation at } X_j}}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

For normalization (正規化) Total sum of squares (總變異量)
For standardization (標準化)

- N is the number of spatial units indexed by i and j ;
- x is the variable of interest;
- $\bar{x}$ is the mean of x ;
- w_{ij} are the elements of a matrix of spatial weights with zeroes on the diagonal (i.e., $w_{ii} = 0$);
- and S_0 is the sum of all w_{ij} , $S_0 = \sum_{i=1}^N \sum_{j=1}^N w_{ij}$.



Fundamental Geo-Processing and GIS Analysis

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Types of Attribute Data

- Nominal data (類別資料)
 - Only classification, and no order and quantitative magnitude
 - Gender, religion, land use, etc.
- Ordinal data (序位或級序資料)
 - Meaningful ranking or order, but no quantitative magnitude
 - Education level, etc.
- Interval data (等距或間隔資料)
 - Meaningful and equal difference between values, but no true zero
 - Temperature, elevation, etc.
- Ratio data (比率資料)
 - Meaningful ratio comparisons, and have a true zero
 - Income, population, etc.

**OK, we know there are different types of attribute data.
But how many classes should we use?**

Data Classification (資料分類)

- All classifications have a target audience or end-user in mind.
- The more closely your classification can be made to fit user needs, the more useful it becomes.
- Common operations:
 - Natural break
 - Equal interval
 - Defined interval
 - Manual interval
 - Quantile
 - Geometric interval
 - Standard deviation

Natural Break

- Classes are based on natural grouping inherent in the data.
- The method identifies break points by picking the class break that best group similar values and maximize the differences between classes.
- Natural breaks are data-specific classifications and not useful for comparing multiple maps built from different underlying information.

Natural Break (Jenks)

- This classification in ArcGIS Pro is based on the Jenks Natural Breaks algorithm.
- Jenks algorithm calculates the sum of squared difference (SSD) for every possible break and determine classes to **minimizes within-class sum of squared differences**. For each break:

$$SSD = \sum (x_i - \bar{x})^2$$

- Jenks algorithm uses goodness of variance fit (GVF) to assess classification.

$$GVF = \frac{SDAM - SDCM}{SDAM}$$

SDAM: sum of squared deviation from the array mean

SDCM: sum of squared deviation from the class mean

Contents

Search



Drawing Order

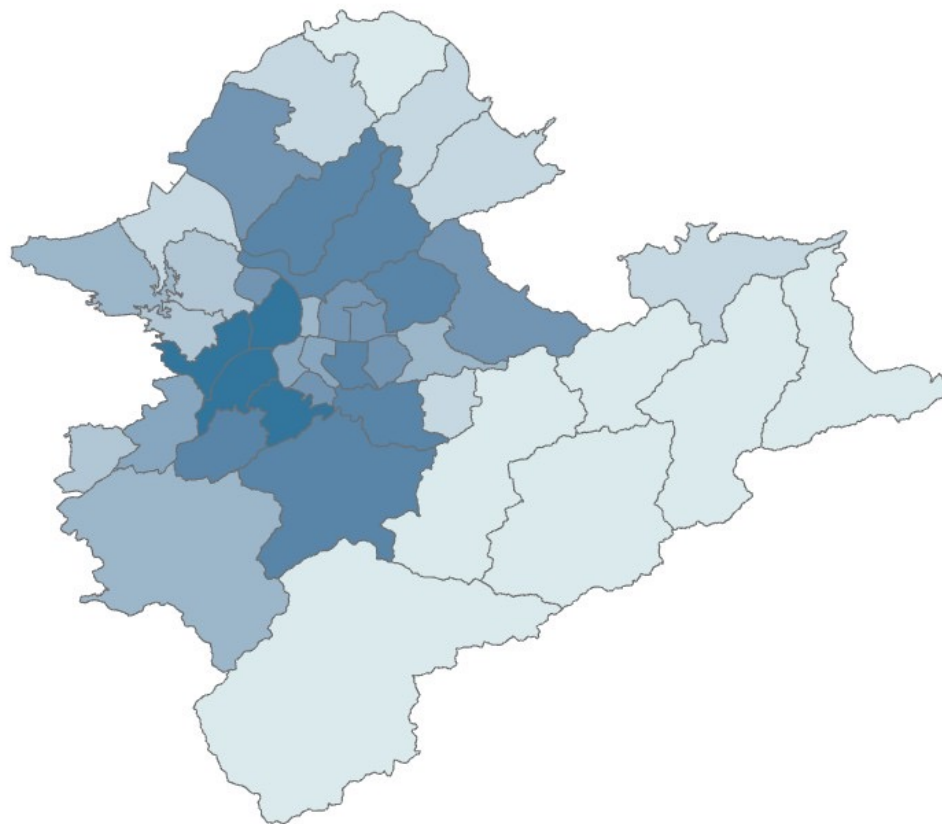
Map

TaipeiMetropolitan_202406pop_...

P_CNT

4095.000000 - 10838.000000
10838.000001 - 42112.000000
42112.000001 - 93474.000000
93474.000001 - 134786.000000
134786.000001 - 179839.000000
179839.000001 - 216082.000000
216082.000001 - 305721.000000
305721.000001 - 554145.000000

Map



Primary symbology

Graduated Colors

Field P_CNT

Normalization <None>

Method Natural Breaks (Jenks)

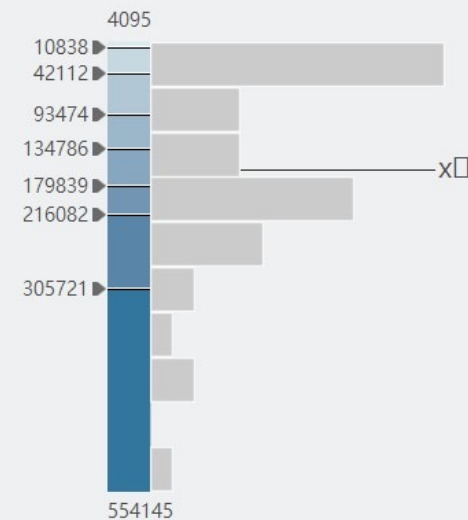
Classes 8

Color scheme

Template

Classes Histogram Scales

More



Equal Interval

- This classification scheme divide the range of attribute values into equal-sized sub-ranges.
- For example, if features have attribute values ranging from 0 to 300 and you have three classes, each class represent a range of 100 with class ranges of 0-100, 101-200, and 201-300.
- Best applied to **familiar data ranges**, such as percentages and temperature.

Contents

Search



Drawing Order

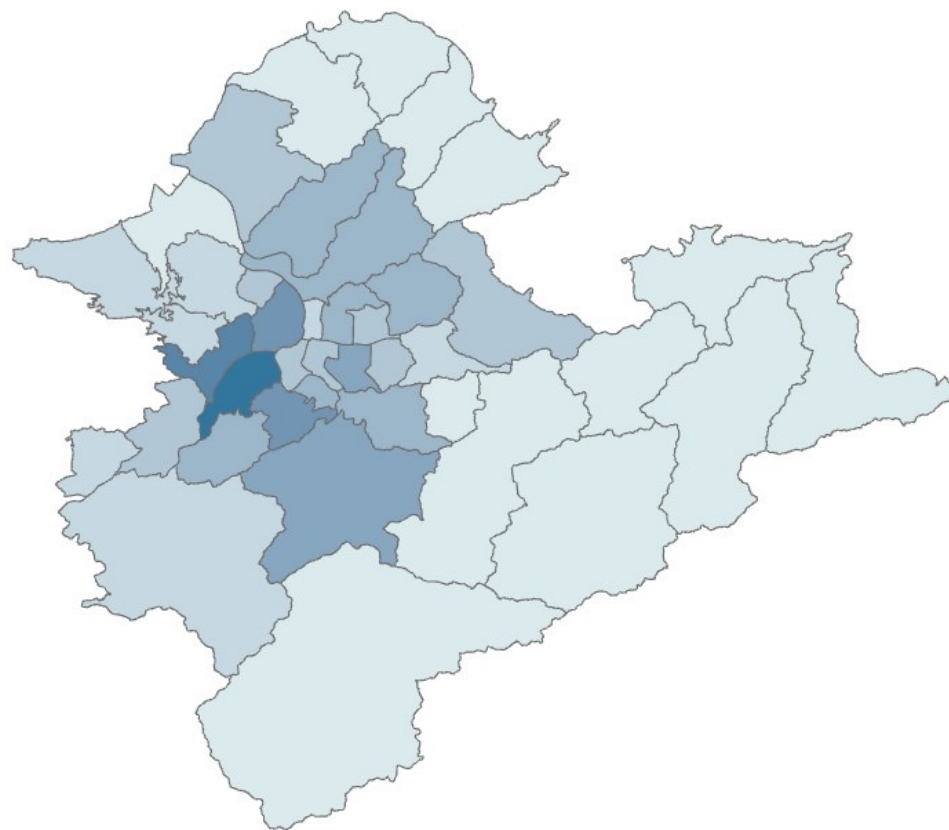
Map

TaipeiMetropolitan_202406pop_...

P_CNT

4095.000000 - 72851.250000
72851.250001 - 141607.500000
141607.500001 - 210363.750000
210363.750001 - 279120.000000
279120.000001 - 347876.250000
347876.250001 - 416632.500000
416632.500001 - 485388.750000
485388.750001 - 554145.000000

Map



Primary symbology

Graduated Colors

Field P_CNT

Normalization <None>

Method Equal Interval

Classes 8

Color scheme

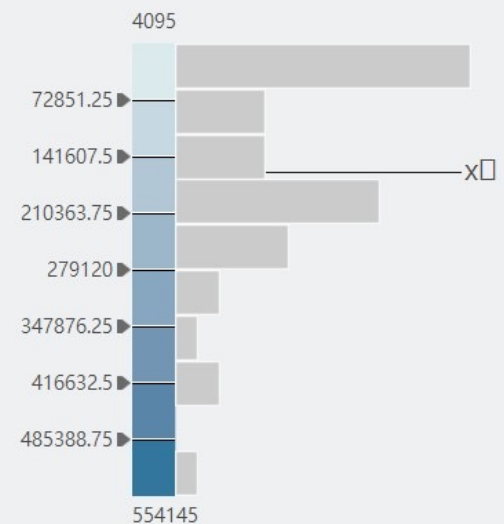
Template

Classes

Histogram

Scales

More



Defined Interval

- This classification scheme allows you to specify an interval by which to equally divide a range of attribute values.
- Rather than specifying the number of intervals as in the equal interval classification scheme, with this scheme, you specify the interval values. ArcGIS automatically determines the number of classes based on the interval.
- For example, if the interval size is 75, each class will span 75 units. The number of classes, based on the interval size and maximum sample size, is determined automatically. The interval size must be small enough to fit the minimum number of classes allowed, which is three.

Contents

Search

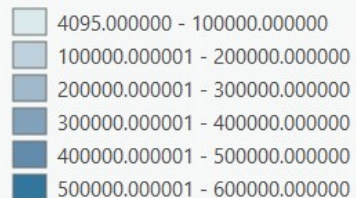


Drawing Order

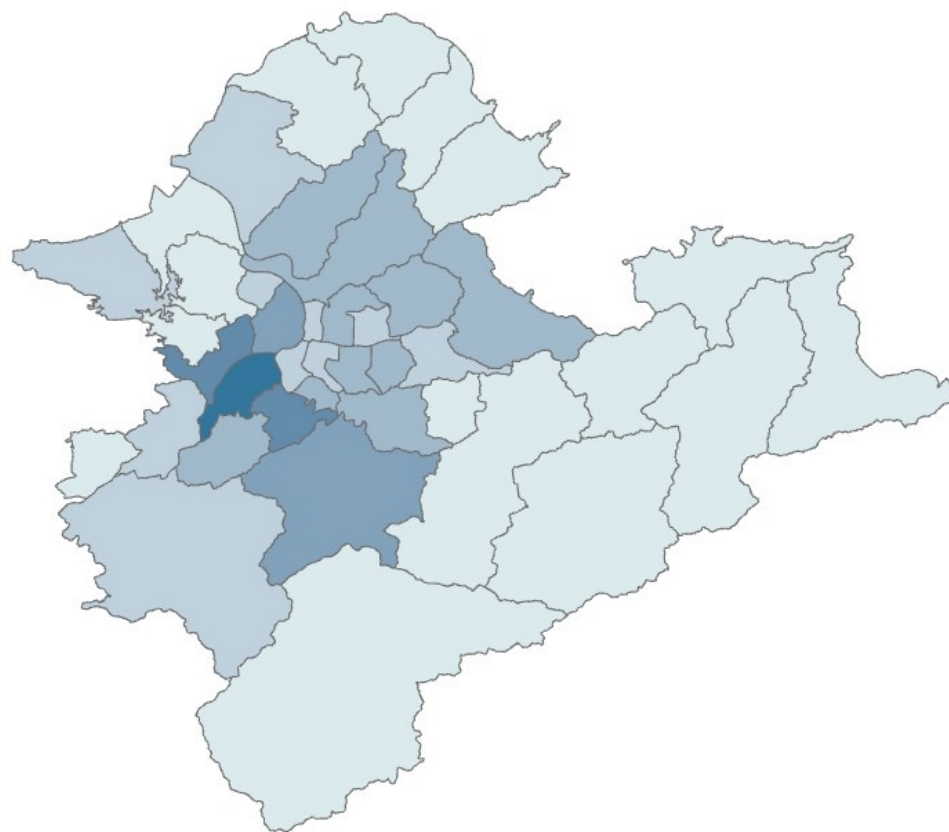
Map

TaipeiMetropolitan_202406pop_...

P_CNT



Map



Primary symbology

Graduated Colors

Field: P_CNT

Normalization: <None>

Method: Defined Interval

Interval size: 100000

Classes: 6

Color scheme: [Blue gradient]

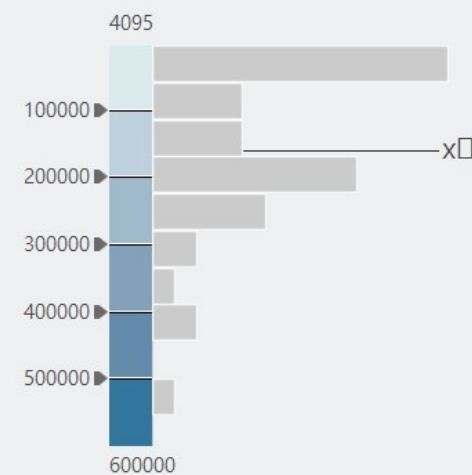
Template: [Orange square]

Classes

Histogram

Scales

More



Manual Interval

- This classification scheme create manually or modify one of the preset classification methods appropriate for your data.

Contents

Search



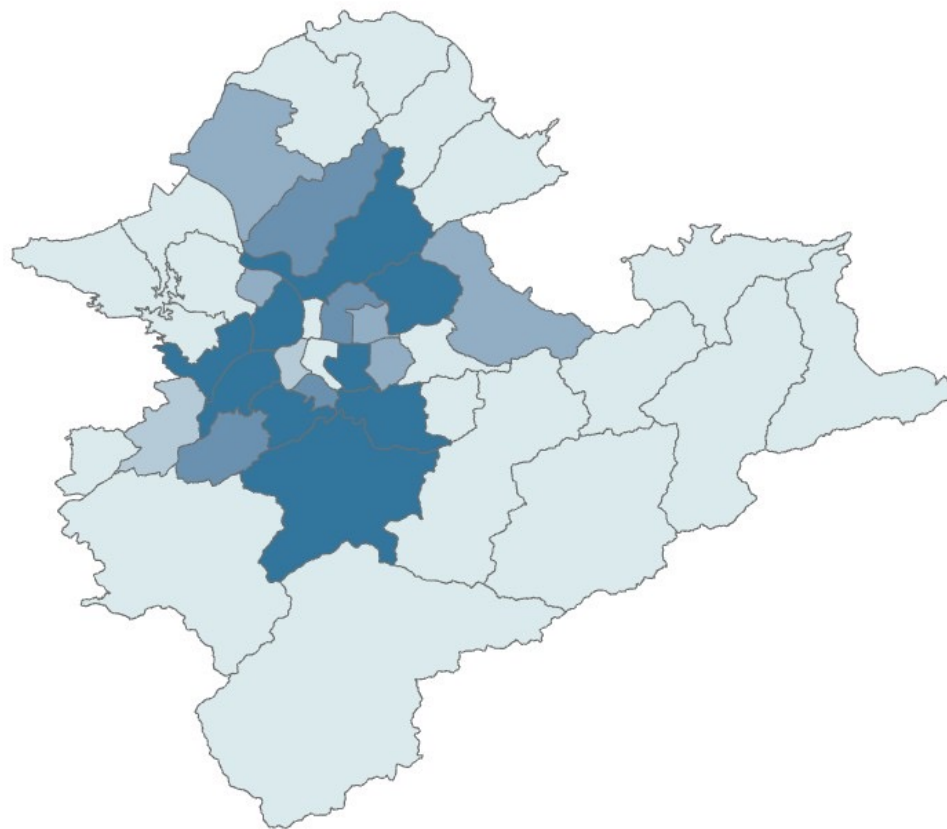
Drawing Order

- Map
- ☒ TaipeiMetropolitan_202406pop_...

P_CNT

- 4100 - 160000
- 170000 - 180000
- 190000 - 210000
- 220000 - 250000
- 260000 - 550000

Map



1:556,119



121.0583671°E 25.38562

Primary symbology

Graduated Colors

Field P_CNT

Normalization <None>

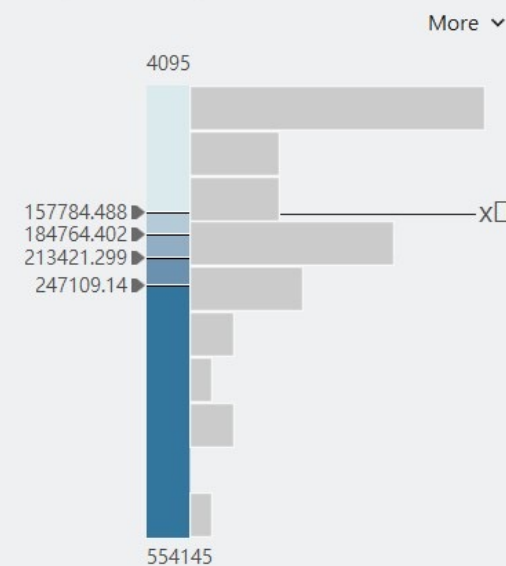
Method Manual Interval

Classes 5

Color scheme

Template

Classes Histogram Scales



Zoom 100%

Bins 10

Quantile

- Each class contains an equal number of features. A quantile classification is well suited to linearly distributed data.
- This method will create a very balanced map, but some of the classes may be close together in terms of their values. You can minimize this distortion by increasing the number of classes.

Contents

Search



Drawing Order

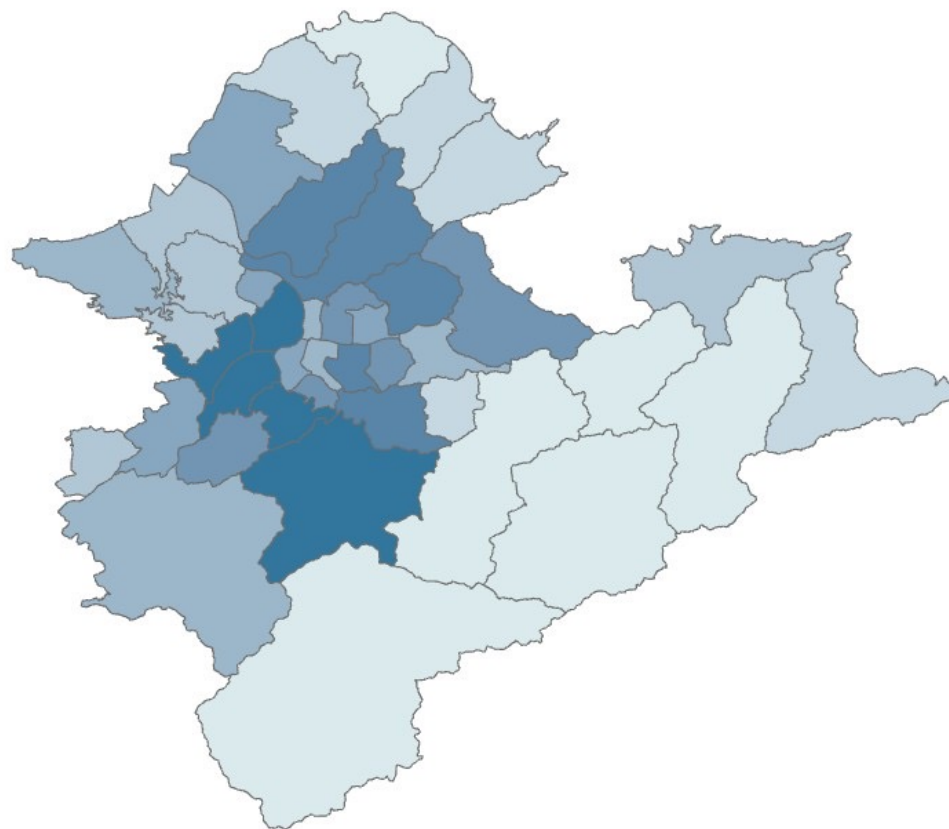
Map

TaipeiMetropolitan_202406pop_...

P_CNT

4095.000000 - 10595.000000
10595.000001 - 23633.000000
23633.000001 - 93474.000000
93474.000001 - 149838.000000
149838.000001 - 199106.000000
199106.000001 - 241006.000000
241006.000001 - 291756.000000
291756.000001 - 554145.000000

Map



Primary symbology

Graduated Colors

Field P_CNT

Normalization <None>

Method Quantile

Classes 8

Color scheme

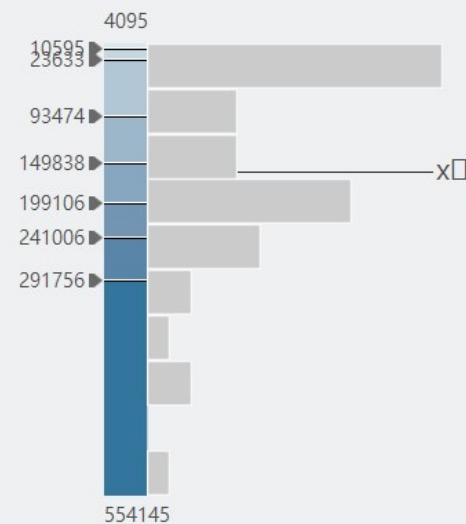
Template

Classes

Histogram

Scales

More



Geometric Interval

- This classification scheme mathematically defined class widths based on a geometric series, giving an approximately equal class width and consistent frequency of observation per class.
- This algorithm was specifically designed to accommodate continuous data.
- One example for using the geometrical interval classification is a rainfall dataset in which only 15 out of 100 weather stations (less than 50 percent) have recorded precipitation, and the rest have no recorded precipitation, so their attribute values are zero.

Contents

Search



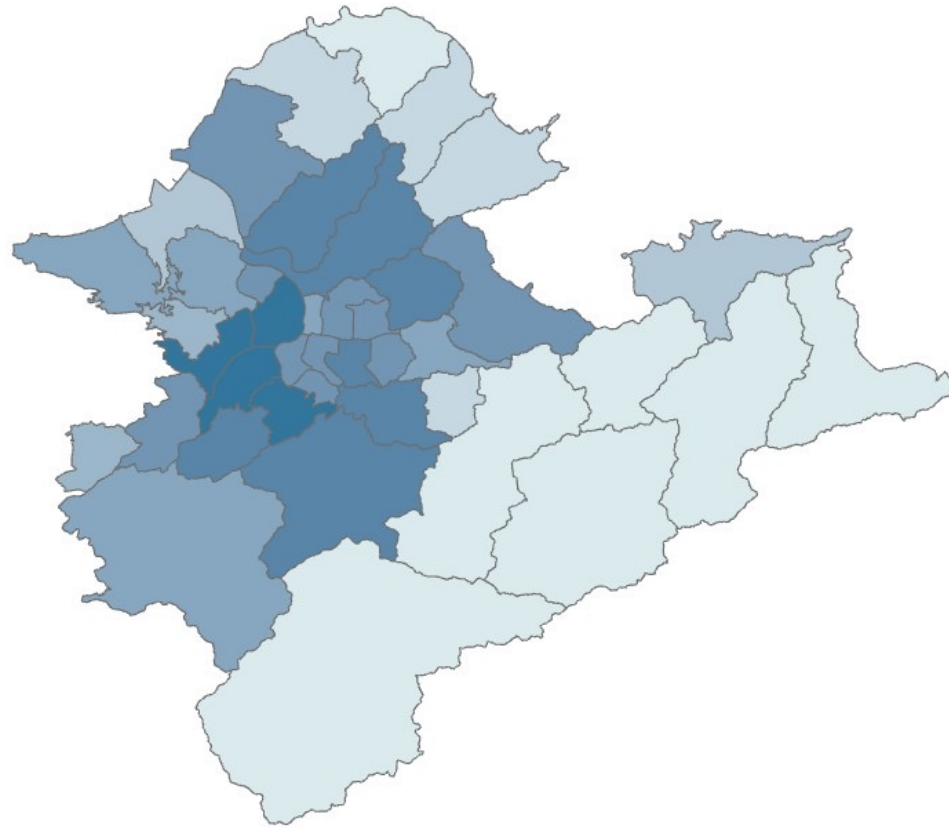
Drawing Order

- Map
- TaipeiMetropolitan_202406pop_...

P_CNT

4095.000000 - 14926.717765
14926.717766 - 31267.346543
31267.346544 - 55918.667290
55918.667291 - 93107.420724
93107.420725 - 149210.028619
149210.028620 - 233845.901617
233845.901618 - 361526.793237
361526.793238 - 554145.000000

Map



1:556,119



121.0583671°E 25.38562

Primary symbology

Graduated Colors

Field: P_CNT

Normalization: <None>

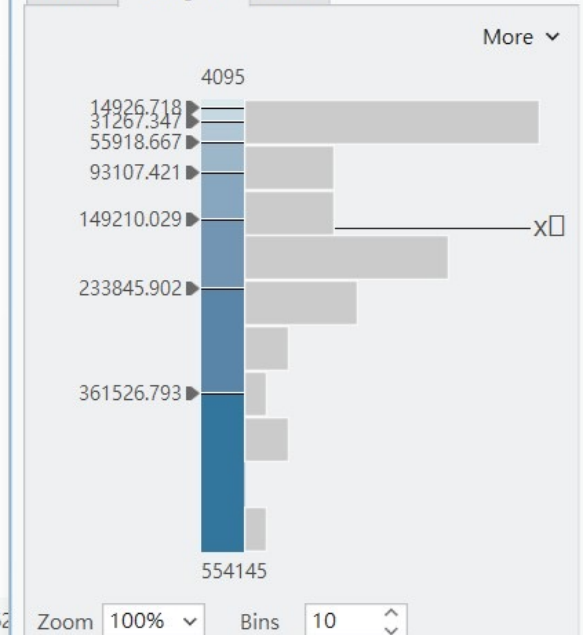
Method: Geometric Interval

Classes: 8

Color scheme: [Blue gradient]

Template: [Orange square]

Classes Histogram Scales



Standard Deviation

- This classification scheme shows you how much a feature's attribute value **varies from the mean**.
- ArcGIS calculates the mean values and the standard deviation from the mean.
- For normally-distributed (常態分布) data, class widths are defined using standard deviations from the mean of the data array, giving an equal class width and varying frequency of observations per class.

Contents

Search



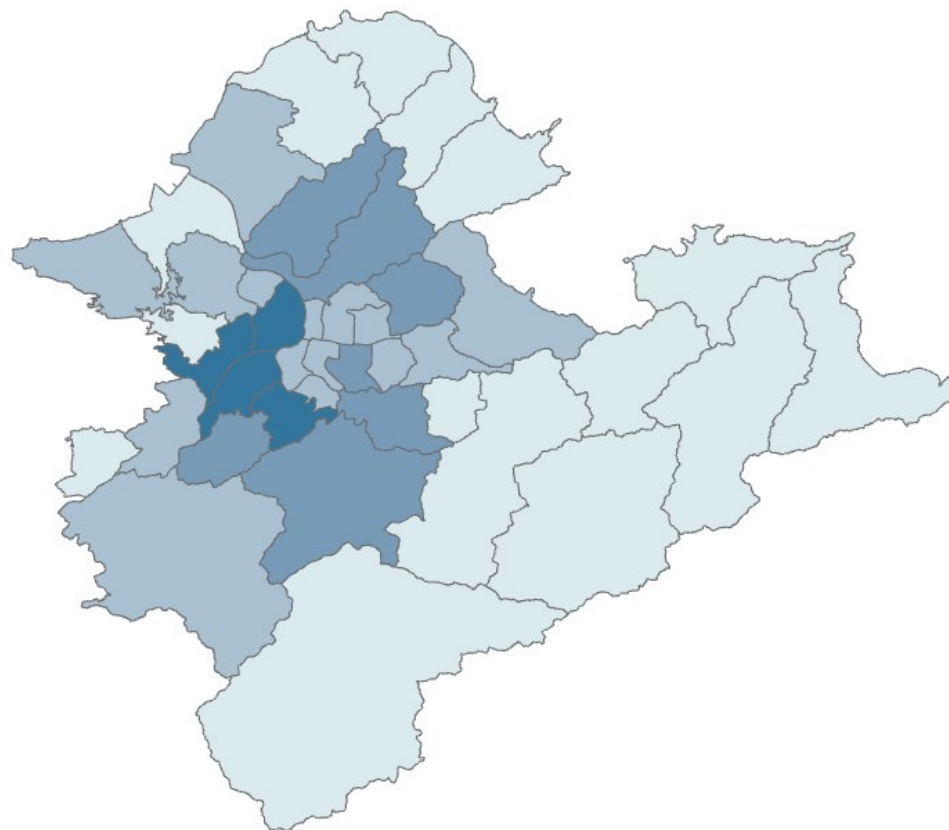
Drawing Order

- Map
- ☒ TaipeiMetropolitan_202406pop_...

P_CNT

- < -0.50 Std. Dev.
- 0.50 - 0.50 Std. Dev.
- 0.50 - 1.5 Std. Dev.
- > 1.5 Std. Dev.

Map



Primary symbology

Graduated Colors

Field: P_CNT

Normalization: <None>

Method: Standard Deviation

Interval size: 1 standard deviation

Classes: 4

Color scheme: [Blue gradient]

Template: [Orange]

Classes

Histogram

Scales

More

